MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, FLEXIBLE, CONVOLUTED, FLUORINATED ETHYLENE PROPYLENE, INTEGRAL BRAID

This specification is approved for use by the Naval Air Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the sleeving described herein shall consist of this document and the issue in effect of MIL-T-81914(AS).

REQUIREMENTS:

Convolution type: Helical (see 3.3)

Construction details: Braided convoluted tubing shall consist of an outer FEP jacket, the braid, and an inner FEP core. Available sizes, dimensions and tolerances, are located on Table I. Lengths shall be as specified by the procuring activity.

Braid: The integral braid shall consist of two layers of nickel coated copper wire, 34 AWG, ASTM B355, Class 4 or equal. Each layer of braid shall have a minimum optical coverage of 85%.

Continuous operating temperature: -67°C (-88°F) to +200°C (392°F)

Color: Unless otherwise specified, the supplied color shall be black.

Physical properties: General physical requirement values along with associated test conditions are located in Table II.

Military part number: Consists of the basic number of this specification sheet and a dash number as shown below:

M81914/7-0000

- Identifier (Table I)
- Class, when applicable (when no classes are specified, the numeric 0 shall be used)
- Type of convolution (1-helical, 2-annular)
- Specification sheet

FSC 9330

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:23Z
Check the source to verify that this is the current version before use.

TABLE I

CONSTRUCTION DETAILS

Identifier	Max. Inside Dia.	Min. Inside Dia.	Max. Outer Dia.	Wall Thickness		Convolutions Per Inch ± 1/2	Weight per 100 Feet, Max.	Min. Bend Radius
				Core Min.	Jacket Max.			
**01	.188	.181	.39	.015	.018	6	10.8	.75
**02	.281	.273	.50	.015	.018	6	13.2	1.00
**03	.312	.303	.54	.015	.018	6	14.6	1.25
**04	.375	.364	.58	.015	.018	6	15.8	1.50
**05	.437	.425	.67	.015	.018	6	17.3	2.00
**06	.500	.485	.71	.015	.018	6	19.3	2.25
**07	.625	.608	.83	.015	.018	6	23.6	2.50
**08	.750	.730	.99	.015	.018	5	26.1	2.87
**09	.875	.850	1.20	.015	.018	5	31.3	3.25
**10	1.000	.975	1.30	.015	.018	5	36.4	3.62
**11	1.125	1.105	1.47	.015	.018	4	42.1	4.00
**12	1.250	1.210	1.61	.015	.018	4	47.9	4.25
**13	1.500	1.440	1.84	.015	.018	4	60.0	4.37
**14	1.750	1.690	2.09	.015	.018	4	75.0	4.62
**15	2.000	1.940	2.34	.015	.018	4	87.0	5.25

** - The asterisks shall be replaced by convolution type and class designation.

Note: Unless otherwise specified all dimensions are in inches.

TABLE II

PHYSICAL PROPERTIES

PROPERTY	REQUIREMENT	TEST METHOD
Construction details	In accordance with Table I	4.6.1
Stress in pounds @ 10% strain	18 to 55	4.6.2
Specific gravity, max.	2.18	4.6.3
Crush resistance, pounds Horizontal, min.	15	4.6.4
Low temperature flexibility	No cracking	4.6.5; 10,000 cycles
Heat shock	No dripping, flowing or cracking	4.6.6; 250 ±2°C (482 ±4°F) for 4 hours
Longitudinal change, %, max.	15	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:23Z
Check the source to verify that this is the current version before use.

TABLE II (Continued)

PROPERTY	REQUIREMENT	TEST METHOD
Heat aging		4.6.7; 225 ±2°C (437 ±4°F) for 96 hours
Stress in lbs at 10% strain, % of initial value	80	
Fluid resistance		4.6.8
Stress in lbs at 10% strain, max. % of original value	80	
Flammability	Non-flammable	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	No breakdown, 12,000 volts	4.6.11

1/ See 6.4, Properties not tested.

Standard procurement: Procured as coils. Unless otherwise specified individual lengths shall be 18 feet.

Preparing Activity
Navy - AS
(Project No. 9330-N752)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-T-81914/7(AS) Tubing, Plastic, Flexible, Convoluted, Fluorinated Ethylene Propylene, Intergral Braid

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:23Z
Check the source to verify that this is the current version before use.